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Case Study 2: Architecture Design of the Proposed Work

Base Paper: CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks. Cheng, T., Li, S., and Pan, Y., Computer Networks, Vol. 287, Art. 112479, 2026

1. Overview

CoWiC-MAC is a coordination framework for WiFi and ZigBee devices that share the 2.4 GHz ISM band. Instead of letting the two technologies contend for the channel on their own, the WiFi Access Point acts as an orchestrator and decides when WiFi transmits and when ZigBee gets interference-free time. Coordination between the two sides happens over a Cross-Technology Communication (CTC) link that carries feedback in both directions. The architecture below shows the components and the flow between them.

2. Architecture Diagram

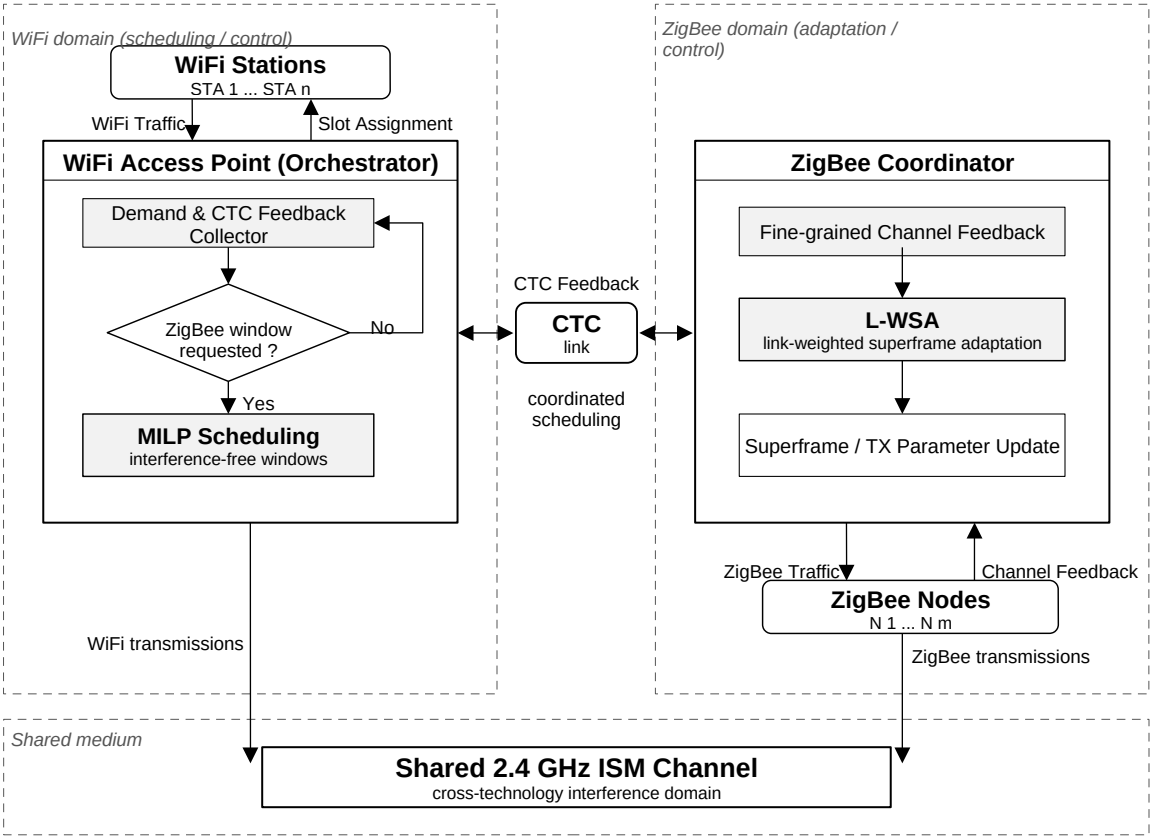


Fig. 1. Architecture of CoWiC-MAC

3. Component Description

WiFi Access Point (Orchestrator). The central component of the design. It has more computational capability than the low-power devices, so the scheduling work is placed here. It decides the order of WiFi transmissions and coordinates with the ZigBee side.

MILP-based Scheduling Model. Runs inside the AP. It uses Mixed-Integer Linear Programming to reorder WiFi traffic so that interference-free transmission opportunities are created for ZigBee. The model also keeps the scheduling overhead bounded, so reordering does not hurt WiFi.

CTC Link (Cross-Technology Communication). The communication path between the WiFi AP and the ZigBee coordinator. WiFi and ZigBee cannot normally talk to each other, so CTC provides the bidirectional feedback that makes coordinated scheduling possible.

ZigBee Coordinator. Manages the ZigBee side of the network. It receives coordination information over the CTC link and applies it to the ZigBee nodes under it.

L-WSA Algorithm (Link-Weighted Superframe Adaptation). A lightweight algorithm running at the ZigBee coordinator. It adapts the transmission parameters using fine-grained channel feedback. It is kept lightweight because ZigBee devices are resource constrained.

WiFi Stations. The normal WiFi client devices. Their traffic is transmitted in the order decided by the MILP schedule at the AP.

ZigBee Nodes. The low-power devices. They transmit during the interference-free windows created by the AP, using the parameters set by L-WSA.

Shared 2.4 GHz ISM Channel. The common medium used by both technologies. This is where cross-technology interference between high-power WiFi and low-power ZigBee occurs, and it is the resource the architecture is coordinating.

4. Data and Control Flow

1. The ZigBee coordinator and the WiFi AP exchange feedback over the CTC link. This makes the coordination demand driven instead of blind contention.

2. The MILP-based scheduling model at the AP reorders WiFi traffic and creates interference-free transmission opportunities for ZigBee, while bounding the scheduling overhead.

3. The AP sends the coordination information to the ZigBee coordinator over the CTC link.

4. The L-WSA algorithm at the ZigBee coordinator adapts the transmission parameters based on fine-grained channel feedback.

5. WiFi stations and ZigBee nodes then transmit on the shared 2.4 GHz ISM channel according to this coordinated schedule, which avoids the collisions caused by uncoordinated access.

5. Mathematical Formulation

The design uses two optimization problems. The WiFi AP solves the first problem. This problem sets the start time of each WiFi frame group (FG). The ZigBee coordinator solves the second problem. This problem sets the superframe in the window that the AP gives.

WiFi Traffic Rescheduling. The AP selects a new start time $s_k^{'}$ for each of the K FGs. The AP makes the value of Equation (1) as small as possible:

$$\begin{aligned} min & \sum_{k=1}^{K} \mathbb{1}\left(\left[s_k^{'},\ s_k^{'}+d_k\right] \cap \left[T_{free}^{start},\ T_{free}^{end}\right] \neq \varnothing\right) \cdot d_k \\ & + \gamma \sum_{k=1}^{K} \mathbb{1}\left(\left[s_k^{'},\ s_k^{'}+d_k\right] \cap \left[T_{sensing}^{start},\ T_{sensing}^{end}\right] \neq \varnothing\right) \cdot d_k \\ & + \lambda \sum_{k=1}^{K} \left|s_k^{'} - s_k\right| + \sum_{k=1}^{K} \mu_k \cdot \mathbb{1}\left(P_k = 1\right) \end{aligned} \quad \#(1)$$

Equation (1) has four terms. The first term is the WiFi airtime in the ZigBee interference-free window. The second term is the WiFi airtime in the sensing window. The weight γ makes this term larger. The third term is the time difference between the new start time and the original start time of each FG. The fourth term is the cost to move an FG to the next cycle.

Equation (2) gives the cost to move an FG to the next cycle:

$$\mu_k = \mu_{base} \cdot \frac{1}{max\left(\epsilon,\ e_k - \left(T_{end} + d_k\right)\right)} \quad \#(2)$$

When the deadline of an FG is near, this cost is high. Thus, the AP does not cause long delays to real-time WiFi traffic.

ZigBee Superframe Adaptation (L-WSA). The ZigBee coordinator selects the slot duration, the number of CAP slots i, the number of CFP slots j and the beacon offset. The coordinator makes the total transmission efficiency in Equation (3) as large as possible:

$$\max_{t_{SfSlot},\ i,\ j,\ t_{offset}} \eta_{total} = m \cdot t_{min} \cdot \eta_{cap}^{p} + \alpha \cdot n \cdot t_{SfSlot} \cdot \eta_{cfp}^{p} \quad \#(3)$$

The weight α makes the CFP term larger, because CFP transmissions have no contention. Equation (4) gives the efficiency of one packet in the CAP and in the CFP:

$$\eta_{cap}^{p} = \frac{t_{pk} - t_{ovh}}{t_{min}}, \qquad \eta_{cfp}^{p} = \frac{t_{pk} - t_{ovh}}{t_{SfSlot}} \quad \#(4)$$

The values m and n count only the slots that are clear of the WiFi fragments $\mathcal{T}^{WiFi}$. These fragments stay in the reserved window:

$$m \& = \sum_{k \in CAP} \mathbb{1}\left(\left[t_k,\ t_k + t_{min} \right] \cap \mathcal{T}^{WiFi} = \varnothing \right) \quad \#(5)$$
$$n \& = \sum_{l \in CFP} \mathbb{1}\left(\left[t_l,\ t_l + t_{SfSlot} \right] \cap \mathcal{T}^{WiFi} = \varnothing \right)$$

The full active period must be in the time gap that the AP gives. It must also be in the slot limits of IEEE 802.15.4:

$$(1 + i + j) \cdot t_{SfSlot} \le dur_{end} - dur_{start}, \qquad 0 \le j \le 7, \qquad i + j \le 15 \quad \#(6)$$

6. Algorithm

The two optimization problems in Section 5 are solved online, by the AP and by the ZigBee coordinator, using the following two algorithms.

Algorithm 1: WiFi Traffic Rescheduling (MILP Scheduling, at the AP). Input: frame groups $\{FG_k\}$, k = 1..K, with original start time τ_k, duration ℓ_k, deadline e_k; ZigBee-free window [T free start, T free end]; sensing window [T sensing start, T sensing end]; weights γ, λ, μ base. Output: new start times $\{\tau_k'\}$ for all K frame groups.

1. Receive the ZigBee window request and the CTC feedback from the ZigBee coordinator.

2. If no ZigBee window is requested, set $\tau_k' = \tau_k$ for every k and go to Step 7.

3. Otherwise, for each FG k, set up the decision variable τ_k' and the indicator terms for overlap with the ZigBee-free window and with the sensing window.

4. Compute μ_k from Equation (2): $\mu_k = \mu\text{base} / \max(\varepsilon, e_k - (\text{Tend} + \ell_k))$ for every FG, using its deadline e_k, so that FGs close to their deadline get a high cost to defer.

5. Solve the MILP to obtain $\{\tau_k'\}$ that minimizes Equation (1): $\min \Sigma_k [I(FG_k' \text{ overlaps the ZigBee-free window}) \cdot \ell_k + \gamma \cdot I(FG_k' \text{ overlaps the sensing window}) \cdot \ell_k + \lambda|\tau_k' - \tau_k| + \mu_k \cdot I(P_k = 1)]$, subject to the K FGs and the non-overlap ordering constraint on the WiFi queue.

6. Identify the interference-free window [T free start, T free end] that remains clear of the rescheduled WiFi traffic and send it to the ZigBee coordinator over the CTC link.

7. Apply $\{\tau_k'\}$ to the WiFi queue and transmit the frame groups in the new order.

Algorithm 2: ZigBee Superframe Adaptation (L-WSA, at the ZigBee Coordinator). Input: interference-free window [dur start, dur end] from the AP; WiFi fragment set T WiFi inside the window, from fine-grained channel feedback; slot limits of IEEE 802.15.4 ($0 \le j \le 7$, $i + j \le 15$); weight α. Output: slot duration t SfSlot, number of CAP slots i, number of CFP slots j, beacon offset t offset.

1. Receive the interference-free window [dur start, dur end] and the CTC feedback from the AP.

2. Sense the channel over the window to build the reserved WiFi fragment set T WiFi.

3. For each candidate (t SfSlot, i, j, t offset), compute m and n from Equation (5): $m = \Sigma_{k \in} \text{CAP } I([t_k, t_k + t\text{min}] \cap \text{TWiFi} = \varnothing)$, $n = \Sigma_{ı \in} \text{CFP } I([t_ı, t_ı + t\text{SfSlot}] \cap \text{TWiFi} = \varnothing)$, the number of CAP and CFP slots clear of T WiFi.

4. Compute η cap and η cfp for the candidate from Equation (4): $\eta cap^{p} = (tpk - tovh)/tmin$, $\eta cfp^{p} = (tpk - tovh)/tSfSlot$.

5. Compute η total for the candidate from Equation (3): $\eta total = m \cdot tmin \cdot \eta cap^{p} + \alpha \cdot n \cdot tSfSlot \cdot \eta cfp^{p}$, with the CFP term weighted by α.

6. Discard the candidate if it violates the slot-limit and window constraints of Equation (6): $(1 + i + j) \cdot tSfSlot \leq durend - durstart$, $0 \leq j \leq 7$, $i + j \leq 15$; otherwise keep it as feasible.

7. After checking all candidates, select the feasible one that maximizes η total.

8. Update the superframe and transmission parameters at the ZigBee coordinator and broadcast them to the ZigBee nodes.

9. Repeat from Step 1 on the next CTC feedback cycle.